

AestimAi and the Universal Comparable Index (UCI): A Cooperative, AI-Assisted Architecture for Non-Monetary Value Measurement

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Abstract

This article introduces AestimAi, a cooperatively governed platform built around the Universal Comparable Index (UCI) — a valuation index measure designed explicitly as a non-monetary instrument that quantifies verified counterperformance (an exchanged object, performed work, delivered energy, or provided service) rather than representing currency, e-money, or a financial instrument under EU law. We describe the rationale for separating measurement from settlement, the three-layer institutional architecture (a technology company holding intellectual property, a cooperative association governing the user-facing platform under one-member-one-vote democracy, and an identity layer based on ISO/IEC 18013-5), and the technical design of an agentic, retrieval-grounded artificial intelligence (AI) valuation engine intended to ground UCI estimates in verified comparable transactions rather than unsupported language-model inference. We further outline a hash-anchoring scheme that provides tamper-evidence for off-chain valuation logs without constituting tokenization, situating the design outside the scope of the EU Markets in Crypto-Assets Regulation (MiCA). Finally, we discuss potential societal benefits — including support for barter and reciprocity-based exchange, financial inclusion, and integration with broader sustainability and energy platforms — alongside open questions for empirical validation and regulatory engagement.

Keywords

cooperative economics; non-monetary value measurement; valuation index; artificial intelligence; retrieval-augmented generation; digital identity; hash-anchoring; MiCA; barter exchange; sustainable development

1. Introduction

Conventional digital marketplaces and financial technologies almost universally couple two functions that are conceptually distinct: the measurement of value and the settlement of a transaction. A price is simultaneously a statement about relative worth and an instruction to transfer money. This coupling has well-documented benefits for liquidity and efficiency, but it also constrains forms of exchange — such as barter, reciprocal service provision, and community-based resource sharing — that may be economically and socially valuable but do not map neatly onto monetary settlement, and that can attract regulatory scrutiny when intermediated digitally at scale.

AestimAi is a platform designed around an alternative premise: that a measurement instrument for value can be separated from any settlement function, and that such an instrument — the Universal Comparable Index, or UCI — can be made rigorous, auditable, and AI-assisted without itself becoming a currency, electronic money, or financial instrument. This article describes the conceptual basis, institutional architecture, technical design of the AI-assisted valuation engine, legal positioning, and prospective societal applications of this approach. The intended audience includes researchers in cooperative and social economics, designers of digital public infrastructure, and policymakers engaged with the regulatory treatment of emerging value-measurement technologies.

2. Background and Motivation

The separation of measurement from settlement is not without precedent. Historical and contemporary local exchange trading systems (LETS), time banks, and mutual credit networks have long used unit-of-account constructs that are deliberately non-transferable or non-convertible to currency, precisely in order to remain outside conventional payment regulation while still enabling meaningful exchange of goods, services, and time. What is comparatively novel in the AestimAi approach is the combination of (i) a formally defined, juridically scoped index measure; (ii) a cooperative ownership and governance structure consistent with International Cooperative Alliance (ICA) principles; (iii) an AI-assisted valuation engine intended to ground estimates in real comparable transactions; and (iv) a cryptographic tamper-evidence mechanism for the resulting data, without recourse to tokenization.

The motivation for this design is twofold. First, there is a practical need for a trustworthy, scalable way to estimate the value of heterogeneous objects, services, and even energy flows for the purposes of barter, donation-in-kind, insurance, inheritance, and circular-economy applications — contexts in which formal markets are thin or absent. Second, there is a structural need to ensure that any such valuation infrastructure does not inadvertently create an unregulated parallel currency or payment system, both for legal compliance and for the integrity of the measurement itself: a valuation index that could be hoarded, speculated upon, or transferred like money would cease to function as a neutral measure.

3. Conceptual Design: The Universal Comparable Index (UCI)

UCI is defined, in the governing statutes of AestimAi Economic Association (the cooperative association), as a valuation index measure — explicitly not a means of payment, a currency, or a financial instrument. Under this definition, UCI always represents a verified counterperformance: a verified exchange of an object, performed work, delivered energy, or a provided service. Pure barter recorded in UCI without an associated, verified counterperformance is not permitted under the statutes.

This definition has several structural consequences. UCI has no transfer function between users: each participant maintains an independent, self-registered log of their own valuations and counterperformances, rather than a shared ledger of balances that could be debited and credited like an account. AestimAi, the cooperative, and UCI itself never become a party to, or intermediary in, any transaction or exchange between users — a principle treated as superordinate to all other provisions in the platform's governing documents, in the sense that it governs the interpretation of every other rule. The role of the platform is therefore limited to measurement, verification support, and record integrity, not to facilitation of transfer or settlement.

4. Institutional Architecture

The platform is organized as a three-layer structure intended to align legal accountability, democratic governance, and technical operation.

- Technology layer: AestimAi Tech AB, a Swedish technology company, holds the intellectual property underlying the AestimAi platform, including the valuation methodology and software, and licenses this exclusively to the cooperative association under a license agreement that fixes a core specification (the technical and definitional core of UCI) as contractually unalterable without the technology company's written consent.
- Governance layer: AestimAi Economic Association is a cooperative association (Economic Association under Swedish law) operating on a strict one-member-one-vote basis, with no differentiated founder or class privileges in its articles of association. The cooperative holds the direct, member-facing relationship with users through a terms-of-service agreement, and is responsible for governance of the UCI definition's application, subject to the immutable core specification held by the technology layer.

- Identity layer: an optional physical identity credential (“AE ID”), based on the ISO/IEC 18013-5 mobile driving licence standard with FIDO2/WebAuthn-based authentication, NFC, and one-time-password fallback, provides a verifiable identity anchor for cooperative membership and for counterperformance verification, without requiring reliance on third-party national identity schemes.

This layered structure is designed to separate (a) who owns and can evolve the underlying technology, (b) who governs its application to a community of users, and (c) how individual users are identified and authenticated — while ensuring that the immutable core definition of UCI cannot be altered by any single layer acting alone.

5. The AI-Assisted Valuation Engine

A central technical challenge for UCI is producing valuations that are both scalable (covering an open-ended range of objects, services, and energy flows) and trustworthy (grounded in evidence rather than unsupported inference). The proposed valuation engine architecture addresses this through eight integrated layers, summarized below.

5.1 Multimodal Perception

The engine accepts photographic and textual descriptions of objects or services to be valued, using a vision-capable large language model to support object identification and condition assessment from images, with media handled through a persistent file-storage interface to avoid repeated uploads across multiple analytical steps.

5.2 Retrieval-Augmented Valuation

Rather than producing an estimate from the language model's internal knowledge alone, the engine grounds each valuation in retrieved comparable transactions (“comparables” or “comps”) drawn from a vector database of structured market data, supplemented by live web search for current market prices. This retrieval-augmented approach is identified as the single largest contributor to valuation accuracy, because it shifts the system from estimation based on generalized patterns to reasoning over verifiable, current evidence — analogous to the methodology of a professional appraiser who relies on comparable sales.

5.3 Agentic Orchestration

A tool-using agent loop structures the valuation process into discrete, auditable steps: identification of the item, retrieval of comparables, application of a statistical (hedonic) pricing model where sufficient data exists, generation of clarifying questions where information is insufficient, fusion of the resulting estimates, and generation of a human-readable justification. Structuring the process in this way supports both robustness and the auditability required for a measurement instrument whose outputs may be hash-anchored for tamper-evidence.

5.4 Ensemble and Probabilistic Fusion

Multiple independent estimators — the language-model estimate, a gradient-boosted hedonic pricing model for data-rich categories, retrieved comparables, and, where available, crowd or respondent estimates — are combined into a posterior distribution using Bayesian fusion. This reduces sensitivity to any single error source, such as an atypical comparable listing or a model overfitted to a particular category, in a manner that is statistically principled rather than a simple average.

5.5 Calibrated Uncertainty via Conformal Prediction

Confidence intervals around each valuation are produced using conformal prediction methods, which provide a statistical coverage guarantee (for example, that a stated interval will contain the true value with

a specified probability across many valuations) regardless of the underlying point-estimate model. For a measurement instrument intended to be both auditable and trustworthy, this represents a substantive improvement over heuristically chosen confidence intervals.

5.6 Adaptive Computation

The engine is designed to allocate computational effort adaptively: more intensive reasoning is applied to difficult or ambiguous valuation cases, and less to straightforward ones, with associated prompt-caching strategies to reduce cost and latency for repeated structural elements such as category taxonomies. Structured output formats ensure that valuation results (point estimate, interval, reasoning, and key contributing factors) are consistently machine-readable for downstream systems.

5.7 Federated and Continual Learning

To improve over time without centralizing raw user data, the architecture envisages a federated learning approach in which local model updates, rather than raw valuation records, are used to refine category-level models, preserving user privacy while allowing the system to adapt to its own user base and data patterns.

5.8 Evaluation, Drift Monitoring, and Anchoring

Ongoing quality assurance is intended to combine automated evaluation (including model-based judging and backtesting against known realized prices) with monitoring for distributional drift over time, and batch processing for large-scale revaluation. Aggregated, anonymized valuation data may be periodically hash-anchored to a public distributed ledger, providing external, tamper-evident verification of the integrity of the valuation record without exposing individual user data and without constituting the issuance of any token or financial instrument.

6. Legal and Regulatory Positioning

A central design constraint is that UCI, as implemented, should not require authorization as a payment institution, electronic money institution, investment firm, or crypto-asset service provider under the relevant EU frameworks — specifically the revised Payment Services Directive (PSD2), the second Electronic Money Directive (EMD2), the Markets in Financial Instruments Directive II (MiFID II), and the Markets in Crypto-Assets Regulation (MiCA). The basis for this positioning rests on several features of the design described above: UCI has no transfer function between user accounts, is not redeemable for, or convertible into, currency or other assets through the platform, always requires an associated verified counterperformance, and is not represented on any distributed ledger as a transferable token — the hash-anchoring mechanism anchors cryptographic digests of off-chain records for tamper-evidence, which is a materially different function from tokenization.

The hash-anchoring approach draws on established standards for trust services and tamper-evident logging, including the eIDAS 2.0 framework's provisions for qualified electronic ledgers, the RFC 3161 time-stamping protocol, relevant ETSI technical specifications for time-stamping policies and permissioned distributed ledgers, and the ISO/IEC 27001 information security management standard. Separately, the platform's tokenized financial instruments — an equity-linked token and an intellectual-property-backed bond instrument issued through a third-party tokenization provider — are explicitly structured as distinct financial instruments under MiFID II and are kept architecturally and contractually separate from UCI, which remains non-tokenized.

This positioning is offered as a design rationale rather than a definitive legal conclusion; formal confirmation through engagement with competent authorities and qualified legal counsel in relevant jurisdictions remains an important next step, and is treated elsewhere in the project's documentation as the subject of a dedicated legal opinion.

7. Cooperative Governance and the ICA Principles

The governance design of AestimAi Economic Association — strict one-member-one-vote voting, the absence of differentiated founder or class privileges, and member-facing terms of service distinct from the technology licensor — is intended to align with the seven principles of the International Cooperative Alliance, namely voluntary and open membership, democratic member control, member economic participation, autonomy and independence, education and training, cooperation among cooperatives, and concern for community. This alignment matters for the present discussion in two respects: it situates AestimAi within a long tradition of cooperative enterprise as a vehicle for activities that sit awkwardly within purely commercial or purely public structures, and it provides an institutional safeguard against the immutable UCI definition being altered for the benefit of any single stakeholder group, since changes to the cooperative's application of the definition require democratic member approval within the bounds set by the technology layer's core specification.

8. Prospective Societal Benefits and Applications

While empirical evaluation of the platform remains at an early stage, several categories of potential benefit follow directly from the design choices described above.

- Support for barter and reciprocity-based exchange: by providing a common, AI-assisted valuation reference for heterogeneous objects and services, UCI may reduce the transaction costs associated with bilateral barter, without requiring either party to convert value into currency.
- Financial inclusion and informal-economy participation: individuals or communities with limited access to formal financial services may benefit from a valuation instrument that does not require a bank account, payment license, or currency conversion, while still producing auditable, AI-grounded estimates.
- Circular economy and resource reuse: consistent valuation of used goods, components, and materials may support reuse, repair, and resource-sharing initiatives by making the relative value of exchanged items more transparent.
- Integration with energy and sustainability platforms: because UCI explicitly includes “delivered energy” as a form of counterperformance, the index may serve as a common valuation reference across energy-as-a-service and related sustainability-oriented platforms, an application area of direct relevance to the authors' broader work on solar-to-fuel and energy-as-a-service systems.
- Identity-linked verification without centralization: the IDCoop identity layer, based on an internationally standardized mobile credential format, offers a path to verified counterperformance and cooperative membership that does not depend on a single national identity infrastructure.

9. Discussion and Open Questions

Several open questions warrant further research and development. First, the empirical performance of the AI-assisted valuation engine — particularly the accuracy gains attributable to retrieval-augmented grounding and the calibration quality of conformal-prediction intervals — has not yet been evaluated against realized outcomes at scale, and systematic backtesting against known transaction prices across diverse object categories will be required. Second, while the design rationale for UCI's status outside payment, e-money, financial-instrument, and crypto-asset regulation is grounded in established legal categories, formal regulatory engagement and jurisdiction-specific legal opinions remain necessary, particularly as the platform scales across borders. Third, the cooperative governance model's resilience — its capacity to maintain the immutable core UCI definition and the non-intermediary principle under commercial pressure or rapid growth — is an institutional question that may benefit from comparative study

of other technology-holding-company-plus-cooperative structures. Finally, the interaction between the non-tokenized UCI measurement layer and the platform's separate tokenized financial instruments (an equity token and an intellectual-property-backed bond) will require ongoing governance attention to maintain the architectural and contractual separation on which the regulatory positioning depends.

10. Conclusion

AestimAi proposes a design in which value measurement is deliberately decoupled from settlement, instantiated through the Universal Comparable Index as a non-monetary, counterperformance-anchored valuation measure, governed by a democratic cooperative under an immutable core specification, and supported by an AI-assisted valuation engine designed for evidential grounding, probabilistic calibration, and auditability via cryptographic hash-anchoring rather than tokenization. The approach draws on established traditions in cooperative economics and complementary currencies while introducing a contemporary technical architecture intended to make rigorous, AI-assisted valuation accessible for barter, reuse, and energy-related exchange at scale. Realizing this potential will depend on empirical validation of the valuation engine, formal regulatory confirmation of the legal positioning outlined here, and continued attention to the governance safeguards that keep measurement and settlement institutionally and technically separate.

Acknowledgements

[To be completed — acknowledgements of collaborators, advisors, and institutional support.]

References

- [1] International Cooperative Alliance. Statement on the Co-operative Identity. ICA, [year].
- [2] European Parliament and Council. Directive (EU) 2015/2366 on payment services in the internal market (PSD2).
- [3] European Parliament and Council. Directive 2009/110/EC on electronic money institutions (EMD2).
- [4] European Parliament and Council. Directive 2014/65/EU on markets in financial instruments (MiFID II).
- [5] European Parliament and Council. Regulation (EU) 2023/1114 on markets in crypto-assets (MiCA).
- [6] European Parliament and Council. Regulation (EU) 2024/1183 amending Regulation (EU) No 910/2014 as regards establishing the European Digital Identity Framework (eIDAS 2.0).
- [7] Internet Engineering Task Force. RFC 3161: Internet X.509 Public Key Infrastructure Time-Stamp Protocol (TSP).
- [8] European Telecommunications Standards Institute. ETSI TS 119 312: Electronic Signatures and Infrastructures — Cryptographic Suites.
- [9] International Organization for Standardization. ISO/IEC 18013-5: Personal identification — ISO-compliant driving licence — Part 5: Mobile driving licence (mDL) application.
- [10] International Organization for Standardization. ISO/IEC 27001: Information security, cybersecurity and privacy protection — Information security management systems.
- [11] Authors' own technical and legal architecture documentation for AestimAi, AestimAi Economic Association Stadgar v3.1, and associated agreements [unpublished internal documentation].

Note: This manuscript is a draft prepared for internal review prior to submission. Bracketed items require completion (author affiliations, dates, acknowledgements) and all regulatory conclusions in Section 6 should be confirmed with qualified legal counsel before publication.